# Component List

Smart Ambulance Green Corridor System (single 4-way junction)

Corrected bill of materials, built from your Project Initiation Document, Tech Stack and Design Specification. Gate-IC counts (74HC00/02/04/08/32/20) are recommended quantities; confirm them once your next-state equations are final, and buy one spare of each IC type.

## 1. Digital ICs (74HC family, 5 V)

| Qty | Component | Value / Part No. | Purpose |
|---|---|---|---|
| 1 | 74HC14 | Hex Schmitt inverter | Debounce for REQ_N/E/S/W, CLEAR and RESET (6 gates) |
| 4 | 74HC00 | Quad 2-input NAND | 2 chips = the four NAND SR request latches (8 gates); 1 for lamp green gates; 1 for next-state logic |
| 1 | 74HC02 | Quad 2-input NOR | Red lamp lines (R = NOR of G and Y for each direction) |
| 1 | 74HC04 | Hex inverter | Yellow lamp lines, GS to request-pending, polarity fixes |
| 2 | 74HC08 | Quad 2-input AND | Next-state logic, counter load logic |
| 2 | 74HC32 | Quad 2-input OR | Next-state logic; CLEAR OR timeout to reset the served latch |
| 1 | 74HC20 | Dual 4-input NAND | Counter terminal-count decode (1, 2, 8, 10 s) |
| 5 | 74HC74 | Dual D flip-flop | 10 flip-flops: 3 FSM state bits, 2 direction register, 4 request synchronizers, 1 CLEAR synchronizer |
| 1 | 74HC148 | 8-to-3 priority encoder | N to I7, E to I6, S to I5, W to I4; I0-I3 tied high; EI tied low |
| 1 | 74HC139 | Dual 2-to-4 decoder | Half A: routes CLEAR to the served direction latch. Half B: EMERG direction to green lines |
| 1 | 74HC138 | 3-to-8 decoder | Decodes FSM state (S0-S5) to lamp-logic select lines |
| 1 | 74HC161 | 4-bit synchronous counter | Timing counter (max count 10 fits in 4 bits); use synchronous load to reset |
| 1 | TLC555 (or NE555) | Timer, 8-pin DIP | 1 Hz astable system clock; see note on NE555 output level |
| 2 | ULN2803A | 8-channel driver, 18-pin | Drives the 12 LEDs (12 of 16 channels used). Required, not optional |

## 2. Clock Circuit (1 Hz, 555 astable)

| Qty | Component | Value / Part No. | Purpose |
|---|---|---|---|
| 1 | Resistor | 1 k, 1/4 W | R1 (555 pin 7 to VCC) |
| 1 | Resistor | 47 k, 1/4 W | R2 fixed part |
| 1 | Trimmer pot | 50 k | R2 adjustable part; sets exactly 1 Hz (f = 1.44 / ((R1 + 2 x R2) x C)) |
| 1 | Capacitor | 10 µF, 16 V electrolytic | Timing capacitor C |
| 1 | Capacitor | 10 nF ceramic | 555 control pin (pin 5) to GND |
| 1 | Resistor | 4.7 k, 1/4 W | Output pull-up to 5 V (needed only if you use NE555) |

## 3. Switch Inputs

| Qty | Component | Value / Part No. | Purpose |
|---|---|---|---|
| 4 | Push button | Momentary, normally open | REQ_N, REQ_E, REQ_S, REQ_W |
| 1 | Push button | Momentary, normally open | CLEAR |
| 1 | Push button (or toggle) | Momentary, normally open | RESET |

| Qty | Component | Value / Part No. | Purpose |
|---|---|---|---|
| 6 | Resistor | 10 k, 1/4 W | Pull-down on each input; also the R of the RC debounce |
| 6 | Capacitor | 1 µF ceramic or film | C of the RC debounce (about 10 ms) |

## 4. Output Stage (Signals)

| Qty | Component | Value / Part No. | Purpose |
|---|---|---|---|
| 4 | LED, 5 mm, red | Standard | Red lamps N, E, S, W |
| 4 | LED, 5 mm, yellow | Standard | Yellow lamps N, E, S, W |
| 4 | LED, 5 mm, green | Standard | Green lamps N, E, S, W |
| 12 | Resistor | 330 ohm, 1/4 W | LED current limiting (about 9 mA each) |

## 5. Power

| Qty | Component | Value / Part No. | Purpose |
|---|---|---|---|
| 1 | Wall adapter | 9 V DC, 500 mA or more | Source |
| 1 | DC barrel jack | Panel or breadboard mount | Adapter input |
| 1 | Toggle switch | SPST | Main power switch |
| 1 | 7805 regulator | TO-220, 5 V, 1 A | 5 V rail (no heatsink needed at this load) |
| 1 | Capacitor | 0.33 µF ceramic | 7805 input |
| 25 | Capacitor | 0.1 µF ceramic | 7805 output (1) + one decoupling cap per IC (24) |
| 1 | Capacitor | 100 µF, 16 V electrolytic | Bulk decoupling on the 5 V rail |
| 1 | LED + 330 ohm resistor | Any color | Power-on indicator (optional) |

## 6. Build Hardware and Tools

| Qty | Component | Value / Part No. | Purpose |
|---|---|---|---|
| 17 | IC socket | DIP-14 | 74HC00 x4, 02, 04, 08 x2, 32 x2, 14, 20, 74 x5 |
| 4 | IC socket | DIP-16 | 74HC138, 139, 148, 161 |
| 2 | IC socket | DIP-18 | ULN2803A |
| 1 | IC socket | DIP-8 | 555 |
| 3 | Breadboard | 830 tie-point | Prototyping, block by block |
| 1 | Perfboard | Large (about 15 x 20 cm) | Final soldered build |
| 1 | Jumper wire kit + 22 AWG solid wire | Assorted | Wiring |
| 1 | Road-layout board | Foam board or plywood | Toy-scale junction with switch panel |
| - | Tools | Multimeter, logic probe, oscilloscope (if available), soldering iron and solder | Testing and assembly |

## 7. Corrections Made to Your Documents

• **LED drive:** 74HC outputs are rated for about 4 mA, but 330 ohm LEDs draw about 9 mA. The ULN2803 is required (2 chips), not optional.

• **Latch count:** four NAND SR latches need 8 NAND gates, which is two 74HC00 chips, not one.

• **Button polarity:** a pull-up plus an inverting Schmitt gate gives a high pulse on press, but the NAND latch needs a low pulse. Use a pull-down (button to +5 V) so the inverter output goes low when pressed.

• **Direction register:** add 2 flip-flops to hold the winning direction when EMERG starts. Without them, a higher-priority press during EMERG changes the green mid-emergency and clears the wrong latch.

• **Timeout clears the latch:** feed the 74HC139 enable from CLEAR OR timeout. Otherwise, after a timeout the served latch is still set and the FSM re-enters EMERG for the same request.

• **74HC148 wiring:** it has active-low inputs and outputs. Map N, E, S, W to I7, I6, I5, I4 (highest input wins). The A1, A0 outputs then read N=00, E=01, S=10, W=11 directly; GS goes low when any request is pending.

• **State flip-flops:** 6 states plus the split ALL_RED (NS-next or EW-next) gives 7 states, so 3 flip-flops.

• **NE555 output level:** at 5 V its high level (about 3.3 V) is marginal for 74HC inputs. Use a CMOS TLC555 or add the 4.7 k pull-up.

• **Decoders:** your docs list 74HC138/139 as one block. The design needs both: a 138 for state decode and a 139 for clear routing and emergency direction decode.

• **CLEAR synchronizer:** CLEAR is asynchronous to the 1 Hz clock, so it needs its own D flip-flop before reaching the FSM.

**Totals:** 24 ICs (21 logic, 1 timer, 2 drivers), 6 push buttons, 12 LEDs. Estimated supply current is well under 100 mA.